

Fine-Tuning FLAN-T5-Small Using LoRA on DialogSum

DATA 266 Assignment Report

SID4: 5996

1. Objective

The objective of this assignment was to fine-tune the pretrained google/flan-t5-small sequence-to-sequence model for dialogue summarization using Low-Rank Adaptation (LoRA). The DialogSum dataset was used with dialogue as the input and summary as the target. The assignment also compared a pretrained baseline against the LoRA fine-tuned model and evaluated two LoRA ranks, $r=4$ and $r=16$.

2. Environment and Dataset

- Execution environment: Google Colab
- GPU: NVIDIA Tesla T4
- Random seed / SID4: 5996
- Base model: google/flan-t5-small
- Dataset: DialogSum
- Dataset split sizes observed: train = 1,999; validation = 499; test = 499
- Fine-tuning subset: 1,000 training samples and 200 validation samples

3. Data Preparation

The dataset was loaded and processed so that each dialogue was used as the model input and its associated human-written summary was used as the target. Tokenization produced input_ids, attention_mask, and labels. Two genuinely different test dialogues were selected for the final inference comparison using test indices 0 and 3.

4. Baseline Inference

Before fine-tuning, the pretrained FLAN-T5-small model was evaluated on the two final test dialogues. The first example concerned an office memo prohibiting Instant Messaging, while the second concerned a traffic jam and recommendations to use public transportation.

Example	Reference Summary	Baseline Output
1	Ms. Dawson helps #Person1# write a memo informing employees that Instant Messaging should no longer be used.	#Person2#: Thank you, sir.
2	#Person2# arrives late because of a traffic jam, and #Person1# recommends public transportation for health and environmental reasons.	Taking the subway would be a lot less stressful than driving.

5. LoRA Configuration

PEFT LoRA adapters were attached to the q and v projection modules of FLAN-T5-small. The main experiment used $r=4$, and the rank experiment used $r=16$. Both configurations used LoRA alpha 16 and dropout 0.05.

Configuration	Rank	Alpha	Dropout	Trainable Params	Trainable %
LoRA $r=4$	4	16	0.05	172,032	0.2230%
LoRA $r=16$	16	16	0.05	688,128	0.8862%

6. Fine-Tuning Results

The final stable training runs completed for three epochs. An earlier $r=16$ attempt produced 0.000000 training loss and NaN validation loss and was discarded. The final stable $r=16$ run used mixed precision disabled (fp16=False, bf16=False).

6.1 LoRA $r=4$

Epoch	Training Loss	Validation Loss
1	1.627956	1.453069
2	1.639476	1.413707
3	1.520933	1.402227

Reported final training loss: 1.6972. Final validation loss: 1.402227.

6.2 LoRA $r=16$

Epoch	Training Loss	Validation Loss
1	1.630945	1.451440
2	1.635241	1.410672
3	1.519345	1.398435

Final validation loss: 1.398435.

7. Post-Fine-Tuning Inference

The same two final test dialogues were used for post-fine-tuning inference. For these examples, $r=4$ and $r=16$ produced essentially identical generated summaries.

Example	LoRA Output ($r=4$ and $r=16$)	Observation
1	#Person1# asks #Person2# to take a dictation for #Person2#. The statement was then repeated multiple times.	The model recognized the dictation context, but the output was highly repetitive and did not fully capture the memo's purpose.
2	#Person2# got stuck in traffic again. #Person2# thinks it would be better if #Person2# started taking public transport system to work. #Person2# thinks it would be better if #Person2# started biking to work.	The output captured the traffic problem and alternative transportation suggestions, although speaker attribution was imperfect.

8. Baseline vs. Fine-Tuned Comparison

For Example 1, the pretrained baseline output ('#Person2#: Thank you, sir.') was largely irrelevant to the requested summary. After LoRA fine-tuning, the model recognized that the conversation involved taking a dictation, but it repeated the same sentence and did not fully capture the policy change regarding Instant Messaging.

For Example 2, the baseline captured only the subway recommendation. The LoRA model included more dialogue-specific context, including the traffic problem and recommendations for public transportation and biking. Thus, fine-tuning improved the amount of relevant information captured, while repetition and speaker-attribution errors remained.

9. LoRA Rank Experiment: $r=4$ vs. $r=16$

Rank	Trainable Parameters	Final Validation Loss	Observed Output Quality
$r=4$	172,032	1.402227	Same observed summaries as $r=16$ on the two final examples.
$r=16$	688,128	1.398435	Slightly lower validation loss, but no noticeable output-quality improvement on the two final examples.

Increasing the LoRA rank from 4 to 16 increased the number of trainable parameters by 4x. The $r=16$ model achieved a slightly lower final validation loss (1.398435 vs. 1.402227), but the generated summaries for the two selected test dialogues were essentially identical. Therefore, the higher rank increased parameter cost without a noticeable qualitative gain for the evaluated examples.

10. Debugging and Training Stability

During experimentation, an $r=16$ training run produced 0.000000 training loss and NaN validation loss for all three epochs. That run was treated as invalid and was not used for the reported metrics. The model and trainer were recreated, the tokenized data was verified to contain valid target tokens, and mixed precision was disabled. The subsequent training run produced finite losses across all epochs.

11. Saved Artifacts

- Executed notebook: LoRA_DialogSum_Assignment.ipynb
- LoRA $r=4$ adapter: dialogsum_lora_r4_adapter.zip
- LoRA $r=16$ adapter: dialogsum_lora_r16_adapter.zip
- RUN_LOG.txt
- METRICS.md
- AI_USE.md

12. Conclusion

LoRA successfully enabled parameter-efficient fine-tuning of FLAN-T5-small for DialogSum. The fine-tuned model captured more dialogue-specific information than the pretrained baseline on the selected examples, particularly for the traffic-related dialogue. However, the memo example exposed repetition and incomplete semantic coverage. Increasing the rank from $r=4$ to $r=16$ reduced validation loss only slightly and did not change the observed generated summaries, showing that a larger LoRA rank did not provide a noticeable qualitative benefit in this small experiment.

Appendix A. Training and Saving Commands

Training

```
train_result_r4 = trainer_r4.train()
```

```
train_result_r16 = trainer_r16.train()
```

Saving adapters

```
lora_model_r4.save_pretrained("./dialogsum_lora_r4_adapter")
```

```
lora_model_r16.save_pretrained("./dialogsum_lora_r16_adapter")
```